LICHTENBAUM'S THEOREM: VANISHING OF TOP COHOMOLOGY IS EQUIVALENT TO NO TOP DIMENSIONAL COMPONENT BEING PROPER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G5F (sources/stacks-duality/pinned-original-source.tex line 9381).

Allowed standard prerequisites (editorial).

- The Lichtenbaum lemma bounding the dimension from nonvanishing top cohomology. Same chapter, Lemma labelled lemma-lichtenbaum, cited at proof lines 9424-9426. Turns nonvanishing of H^d on an integral subscheme into dimension d and properness.
- Dualizing modules for a proper variety over a field. Same chapter, Lemma labelled lemma-duality-proper-over-field, cited at proof lines 9396-9400. Produces omega_{X'} with H^d(X', omega_{X'}) nonzero.
- Filtrations of coherent modules with integral quotients. Cohomology of Schemes, Lemma labelled coherent-lemma-coherent-filter, cited at proof lines 9410-9412. Reduces the coherent case to a single integral closed subscheme with an ideal sheaf.
- Quasi-coherent modules as filtered colimits of coherent submodules and vanishing of their cohomology. Properties, Lemma labelled properties-lemma-quasi-coherent-colimit-finite-type, cited at proof lines 9430-9434, and Cohomology, Lemma labelled cohomology-lemma-quasi-separated-cohomology-colimit, cited at proof lines 9434-9436. Proves the equivalence of (1) and (2).

Source reference. Original lemma 0FVV: sources/stacks-duality/pinned-original-source.tex, source lines 6934–6934. Same-chapter pointer into the author environment lemma at source line 6934; invoked at proof line 9398 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01YF: sources/stacks-chow/tags.txt, source lines 2554–2554. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9412 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G5E: sources/stacks-duality/pinned-original-source.tex, source lines 9329–9329. Same-chapter pointer into the author environment lemma at source line 9329; invoked at proof line 9424 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01PG: sources/stacks-chow/tags.txt, source lines 2229–2229. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9434 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01FF: sources/stacks-chow/tags.txt, source lines 1868–1868. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9436 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Theorem 0.1. *Let X be a nonempty separated scheme of finite type over a field k. Let $d = \dim(X)$. The following are equivalent*

1. *$H^d(X, \mathcal{F}) = 0$ for all coherent $\mathcal{O}_X$-modules $\mathcal{F}$ on X,*
2. *$H^d(X, \mathcal{F}) = 0$ for all quasi-coherent $\mathcal{O}_X$-modules $\mathcal{F}$ on X, and*
3. *no irreducible component $X' \subset X$ of dimension d is proper over k.*

Proof. Assume there exists an irreducible component $X' \subset X$ (which we view as an integral closed subscheme) which is proper and has dimension d. Let $\omega_{X'}$ be a dualizing module of X' over k, see Lemma 0FVV. Then $H^d(X', \omega_{X'})$ is nonzero as it is dual to $H^0(X', \mathcal{O}_{X'})$ by the lemma. Hence we see that $H^d(X, \omega_{X'}) = H^d(X', \omega_{X'})$ is nonzero and we conclude that (1) does not hold. In this way we see that (1) implies (3).

Let us prove that (3) implies (1). Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module such that $H^d(X, \mathcal{F})$ is nonzero. Choose a filtration

$$0 = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_m = \mathcal{F}$$

as in Cohomology of Schemes, Lemma 01YF. We obtain exact sequences

$$H^d(X, \mathcal{F}_i) \to H^d(X, \mathcal{F}_{i+1}) \to H^d(X, \mathcal{F}_{i+1}/\mathcal{F}_i)$$

Thus for some $i \in \{1, \ldots, m\}$ we find that $H^d(X, \mathcal{F}_{i+1}/\mathcal{F}_i)$ is nonzero. By our choice of the filtration this means that there exists an integral closed subscheme $Z \subset X$ and a nonzero coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_Z$ such that $H^d(Z, \mathcal{I})$ is nonzero. By Lemma 0G5E we conclude $\dim(Z) = d$ and Z is proper over k contradicting (3). Hence (3) implies (1).

Finally, let us show that (1) and (2) are equivalent for any Noetherian scheme X. Namely, (2) trivially implies (1). On the other hand, assume (1) and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then we can write $\mathcal{F} = \operatorname{colim} \mathcal{F}_i$ as the filtered colimit of its coherent submodules, see Properties, Lemma 01PG. Then we have $H^d(X, \mathcal{F}) = \operatorname{colim} H^d(X, \mathcal{F}_i) = 0$ by Cohomology, Lemma 01FF. Thus (2) is true. □